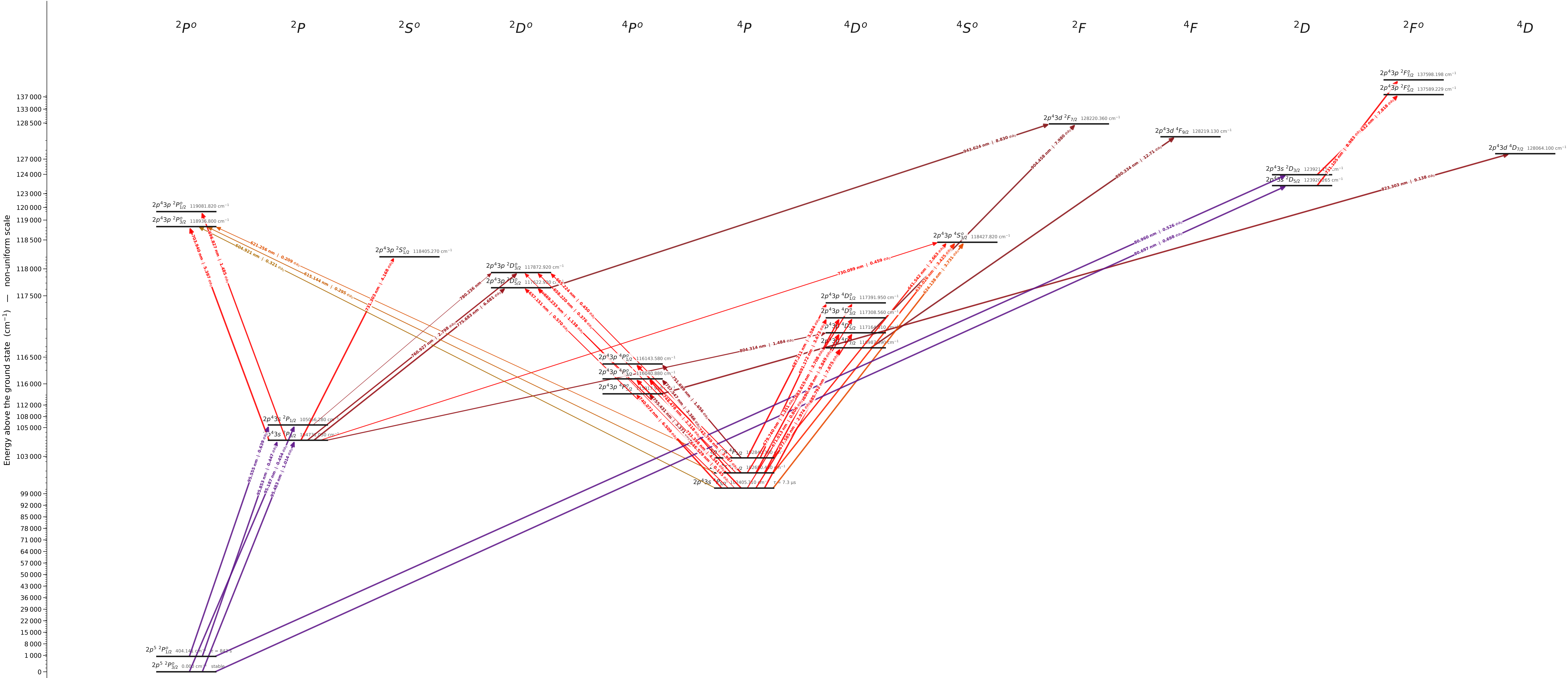
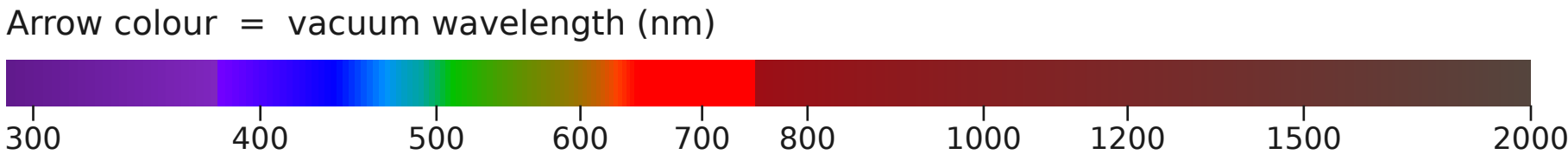


Fluorine

F (F I). Strongest classified transitions below 2 μm (NIST): 27 levels, 46 lines.

F⁺ ionisation limit 140524.50 cm⁻¹ = 17.42282 eV (71.162 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| e r \| J' \rangle |$ in ea0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea0)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
2p ⁴ 3s ⁴ P _{3/2} - 2p ⁴ 3p ⁴ D _{3/2} ^o	677.5848	677.3978	442.4427	2.974	9.600e+06	-	-	677.4 nm
2p ⁴ 3s ⁴ P _{3/2} - 2p ⁴ 3p ⁴ D _{5/2} ^o	685.7918	685.6027	437.14789(15)	7.875	4.870e+07	-	-	685.6 nm
2p ⁴ 3s ² P _{3/2} - 2p ⁴ 3p ² D _{3/2} ^o	760.9267	760.7173	393.9833	2.798	9.000e+06	-	-	761 nm: diode-laser absorption hyperfine spectroscopy
2p ⁴ 3s ² P _{3/2} - 2p ⁴ 3p ² D _{5/2} ^o	775.6827	775.4692	386.4885	6.681	3.230e+07	-	-	775 nm: diode-laser saturated-absorption hyperfine spectr...
2p ⁴ 3s ² P _{1/2} - 2p ⁴ 3p ² D _{3/2} ^o	780.2357	780.0211	384.2332	-	-	-	-	780 nm: diode-laser Doppler-free saturated-absorption hyp...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁹F hyperfine structure (I = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2p ⁵ 2P _{3/2} ^o	2009.990(10)	-	F=1: -2512.49 F=2: +1507.49
2p ⁵ 2P _{1/2} ^o	10244.21(3)	-	F=0: -7683.16 F=1: +2561.05
2p ⁴ 3s ⁴ P _{3/2}	2643.0(10)	-	F=2: -4625.25 F=3: +3303.75
2p ⁴ 3s ² P _{3/2}	3057.9(21)	-	F=1: -3822.38 F=2: +2293.43
2p ⁴ 3s ² P _{1/2}	1737.1(4)	-	F=0: -1302.83 F=1: +434.27
2p ⁴ 3p ⁴ P _{3/2}	1824(20)	-	F=1: -2280.00 F=2: +1368.00
2p ⁴ 3p ⁴ D _{7/2}	1564.0(10)	-	F=3: -3519.00 F=4: +2737.00
2p ⁴ 3p ⁴ D _{5/2}	1148.0(10)	-	F=2: -2009.00 F=3: +1435.00
2p ⁴ 3p ² D _{5/2}	1746.5(15)	-	F=2: -3056.38 F=3: +2183.12
2p ⁴ 3p ² D _{3/2}	1857.1(21)	-	F=1: -2321.38 F=2: +1392.83
2p ⁴ 3d ⁴ D _{7/2}	793(20)	-	F=3: -1784.25 F=4: +1387.75

Measured A, B constants; sources in the data file.